Causal Inference for Intervention & Service Evaluations

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Table of contents

Welcome

This is a practical handbook on causal inference methods for evaluating policy interventions and service changes within the NHS, with a focus on leveraging observational real-world data to generate robust evidence.

The handbook was written initially as part of an NHS England PhD Internship done jointly with the NHS North Central London Integrated Care Board and University College London, and will continue to be periodically updated with revisions following feedback from academics, data scientists, and public health practitioners.

NHS England

NHS North Central London Integrated Care Board

UCL

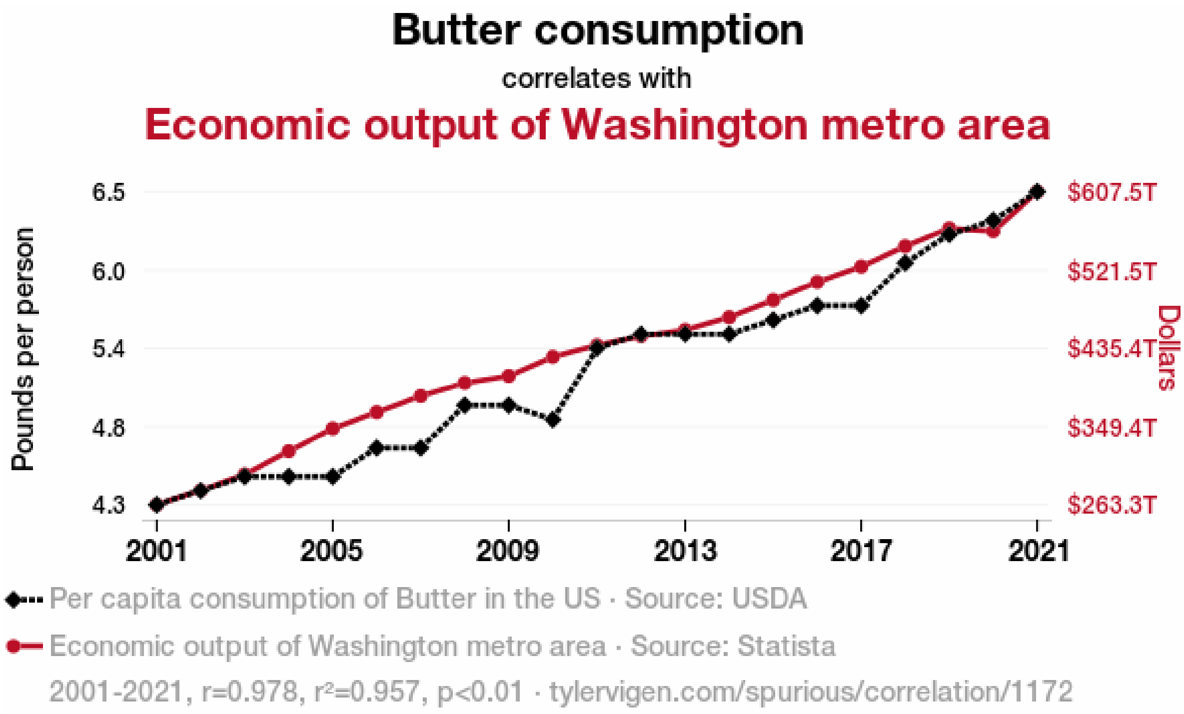
1 Introduction

1.1 Background & Motivation for Causal Inference in Service Evaluations

When performing evaluations of healthcare services or interventions in the NHS, our key evaluation question of interest is almost always causal in nature: • Has our new breast cancer screening programme reduced mortality? • Did implementing a community-based model of care divert patients away from A&E services? • Did implementing a hub and spoke model reduce clinical harm and lower patient waiting times?

Yet many of the methods currently used to evaluate interventions within the NHS are often descriptive in nature. Dashboards are frequently used in this manner, where any changes in an outcome of interest are fully attributed to a single change in implementation in services, without considering whether there are alternative explanations and/or other confounding factors are involved that could instead explain the change instead.

To better illustrate this point, Tyler Vigen publishes great examples of spurious correlations on his webpage (https://tylervigen.com/spurious-correlations), from which I've attached an example of below Vigen (2024):



1.2 A Brief History on Conceptualising Causality

1.3 Common Assumptions in Causal Inference Methodology

1.4 Existing Guidance on Study Design Selection

1.5 New Unifying Guidance on Choosing Causal Inference Methods (Method Selection Tree)

1.6 Key Resources

1.7 Coverage of Causal Inference Methods in Key Texts

Part I

Key Concepts in Causal Inference

TBC

2 Randomized Controlled Trials

TBC

3 Directed Acyclic Graphs

TBC

4 Target Trial Emulation

TBC

5 Defining Treatment Strategies

TBC

Part II

Quasi-Experimental Methods

TBC

6 Difference-in-Differences

TBC

7 Interrupted Time Series

TBC

7.1 Standard Interrupted Time Series Analysis

7.2 Controlled Interrupted Time Series Analysis

8 Synthetic Controls

TBC

9 Regression Discontinuity

TBC

10 Instrumental Variables

TBC

11 Extensions of Quasi-Experimental Designs

TBC

11.1 Negative Controls

11.2 Sensitivity Analysis

Part III

Adjustment-Based Methods

TBC

12 Outcome Regression (Selection on Observables)

TBC

13 Matching (Exact & Covariate)

TBC

14 Propensity Scores

TBC

14.1 Estimating Propensity Scores

14.2 Propensity Score Matching

15 G-Methods: An Introduction

TBC

16 G-Methods 1: Inverse Probability Weighting

TBC

17 G-Methods 2: Parametric G-Formula

17.1 Parametric G-Formula (Non-Iterative Expectation)

17.2 Iterative Conditional Expectation (ICE) G-Formula

18 Extensions of Adjustment-Based Methods

TBC

Part IV

Applying Causal Methods: Worked Examples in the NHS

TBC

19 Case Study 1a: DOAC Head-to-Head Emulated Trial

TBC

20 Case Study 1b: DOAC Policy Prescription Change Interrupted Time Series Analyses

TBC

21 # Case Study 2: Ophthalmology Hub of Care Model Interrupted Time Series Analyses

TBC

22 Practical Barriers to Implementation: Common Challenges in Causal Inference

TBC

23 Conclusion

TBC

24 About the Author & Acknowledgements

24.1 About Me

Hi, my name is Jamie and I'm an MRC funded PhD student at UCL, focusing on the application of causal inference methods within Epidemiology, Data Science, and Health Economics.

Before my PhD, I previously graduated with a BSc in Population Health Sciences (Data Science) from UCL, and an NIHR funded MSc in Epidemiology from the London School of Hygiene and Tropical Medicine.

Prior to my PhD, my interest in researching the application of causal inference methods in observational data stemmed from my varied experience working across multiple areas of public health research, including: • Performing secondary analysis on a nutritional labelling RCT at the NIHR Obesity Policy Research Unit • Supporting cancer trials as a Clinical Trials Assistant in the MRC Clinical Trials Unit • Identifying barriers to STI testing at the NIHR Health Protection Research Unit in Blood Borne and Sexually Transmitted Infections • Quantifying the effects of acute and long-term exposure of air pollutants on respiratory related hospital admissions and related outcomes at UCL • Evaluating the risk of long-term adverse events in cancer survivors using electronic health record data at LSHTM

Having worked with both randomized trials and observational data, focusing my research on causal inference in real-world data felt like a natural next step.

If you have any questions do feel free to reach out to me via the following channels: UCL Email: jamie.wong [at] ucl.ac.uk NHS Email: laichunjamie.wong [at] nhs.net LinkedIn: https://www.linkedin.com/in/jamiewlc/ GitHub: https://github.com/jamiewlc

24.2 Acknowledgements

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Vigen, Tyler. 2024. "Tyler Vigen Spurious Correlation #1,172: Butter Consumption Correlates with Economic Output of Washington Metro Area." TylerVigen.com. January 2024. https://tylervigen.com/spurious/correlation/1172_butter-consumption_correlates-with_economic-output-of-washington-metro-area.